

Lab 8 - Transient Response II

<Your Name>

<Lab Date>

Description

In this lab we study the transient response of a second-order circuit built from an op-amp, two resistors, and two capacitors. The circuit is a unity-gain Sallen-Key low-pass filter, which lets us synthesize a second-order response without an inductor. We design the circuit three ways so that it produces a critically damped, an overdamped, and an underdamped step response, all at a resonant frequency near 1 kHz. For each case we solve the governing differential equation by hand, simulate the circuit in LTSpice, and measure the step response on the bench with a 100 Hz, 2 V peak-to-peak square wave as the input. We then compare the three sets of results.

Procedure

1. We designed three Sallen-Key circuits in LTSpice using the design equations for α , ω_0 , and Q , choosing standard component values that place each circuit in one of the three damping regimes while keeping f_0 near 1 kHz.
2. We drove each simulated circuit with a 100 Hz, 2 V peak-to-peak square wave and recorded the transient response.
3. We built each circuit on the XK700 trainer using a 741 op-amp and the component values from our design table.
4. We used the function generator on the XK700 to apply a 100 Hz, 2 V peak-to-peak square wave to the input.
5. We displayed the input on CH1 and the output on CH2 of the oscilloscope and recorded the waveforms for each of the three cases.

Calculations

Governing equation. Applying KCL at the two internal nodes of the Sallen-Key circuit and combining the results gives the second-order differential equation for the output:

$$R_1 C_1 R_2 C_2 \frac{d^2 V_{out}}{dt^2} + (R_1 + R_2) C_2 \frac{dV_{out}}{dt} + V_{out} = V_{in}.$$

Dividing through by $R_1 C_1 R_2 C_2$ puts this in standard form,

$$\frac{d^2 V_{out}}{dt^2} + 2\alpha \frac{dV_{out}}{dt} + \omega_0^2 V_{out} = \omega_0^2 V_{in},$$

where the damping factor and resonant frequency for this topology are

$$2\alpha = \frac{1}{R_{eq} C_1}, \quad \omega_0^2 = \frac{1}{R_1 C_1 R_2 C_2}, \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}.$$

The quality factor is $Q = \omega_0/(2\alpha)$, so $Q = \frac{1}{2}$ separates the overdamped ($Q < \frac{1}{2}$) and underdamped ($Q > \frac{1}{2}$) cases. At DC the capacitors are open and the follower passes the input straight through, so the DC gain is one and the particular solution is $V_{out} = V_{in}$.

Component selection. We fixed the geometric-mean time constant to hold ω_0 near $2000\pi \text{ rad s}^{-1}$ and adjusted the component ratios to set Q . For the critically damped case we used equal resistors and equal capacitors, which forces $Q = \frac{1}{2}$ exactly. For the overdamped case we kept the capacitors equal and split the resistors, which drops Q below $\frac{1}{2}$ while keeping $\sqrt{R_1 R_2}$ fixed. For the underdamped case we kept the resistors equal and raised C_1 above C_2 , which pushes Q above $\frac{1}{2}$. Table ?? lists the three designs and the resulting parameters.

Table 1: Component values and derived parameters for the three cases.

Case	R_1	R_2	C_1	C_2	Q	f_0 (Hz)	α (s^{-1})
Critically damped	16 k Ω	16 k Ω	10 nF	10 nF	0.500	994.7	6250
Overdamped	47 k Ω	5.6 k Ω	10 nF	10 nF	0.308	981.0	9992
Underdamped	5.1 k Ω	5.1 k Ω	100 nF	10 nF	1.581	986.9	1961

Step response. The square-wave input steps by $\Delta V = 2 \text{ V}$ on each edge. Since the circuit fully settles in well under the 5 ms half-period, every edge produces a complete step response. Both capacitor voltages are continuous through an edge and the circuit carries no DC current in steady state, so the initial conditions after a step are $V_{out}(0) = V_0$ and $\dot{V}_{out}(0) = 0$, where V_0 is the level before the edge. Writing the output as $V_{out}(t) = V_0 + \Delta V g(t)$, the normalized response $g(t)$ rises from 0 to 1 and is found below for each case.

Critically damped ($\alpha = \omega_0 = 6250 \text{ s}^{-1}$, repeated root $s = -6250 \text{ s}^{-1}$):

$$g(t) = 1 - (1 + \alpha t) e^{-\alpha t}.$$

This rises to the final value in the shortest time possible without overshoot, with a time constant $1/\alpha = 160 \mu\text{s}$.

Overdamped (real roots $s_1 = -2128 \text{ s}^{-1}$, $s_2 = -17857 \text{ s}^{-1}$):

$$g(t) = 1 + \frac{s_2 e^{s_1 t} - s_1 e^{s_2 t}}{s_1 - s_2} = 1 - 1.135 e^{-2128 t} + 0.135 e^{-17857 t}.$$

The slow root ($\tau_1 = 470 \mu\text{s}$) dominates, so the output climbs to its final value slowly and without overshoot.

Underdamped (complex roots $s = -1961 \pm j 5882 \text{ s}^{-1}$, $\omega_d = 5882 \text{ rad s}^{-1}$, $f_d = 936 \text{ Hz}$):

$$g(t) = 1 - e^{-\alpha t} \left(\cos \omega_d t + \frac{\alpha}{\omega_d} \sin \omega_d t \right), \quad \frac{\alpha}{\omega_d} = \frac{1}{3}.$$

This overshoots and rings. The predicted overshoot is $e^{-\pi\alpha/\omega_d} = e^{-\pi/3} = 35.1\%$ at a peak time $t_p = \pi/\omega_d = 534 \mu\text{s}$.

Figure ?? plots the three normalized responses computed from the expressions above.

Figure 1: Theoretical normalized step response for the three designs.

Simulations

We built each circuit in LTSpice with the op-amp configured as a unity-gain buffer and drove it with a 100Hz, 2V peak-to-peak square wave. The netlists below use an ideal high-gain buffer for the follower, which reproduces the same transfer function as the 741 in the useful frequency range. The transient plots follow each netlist.

```
* Sallen-Key low-pass, critically damped (Q=0.5, f0~995 Hz)
Vin in 0 PULSE(-1 1 0 1u 1u 5m 10m)
R1 in n1 16k
R2 n1 n2 16k
C1 out n1 10n
C2 n2 0 10n
Eop out 0 n2 out 1Meg ; unity-gain buffer (V+=n2, V-=out)
.tran 0 20m 0 2u
.end
```

Listing 1: Critically damped netlist

<Insert LTSpice transient plot: critically damped>

Figure 2: LTSpice transient response, critically damped case.

```
* Sallen-Key low-pass, overdamped (Q=0.31, f0~981 Hz)
Vin in 0 PULSE(-1 1 0 1u 1u 5m 10m)
R1 in n1 47k
R2 n1 n2 5.6k
C1 out n1 10n
C2 n2 0 10n
Eop out 0 n2 out 1Meg
.tran 0 20m 0 2u
.end
```

Listing 2: Overdamped netlist

<Insert LTSpice transient plot: overdamped>

Figure 3: LTSpice transient response, overdamped case.

```
* Sallen-Key low-pass, underdamped (Q=1.58, f0~987 Hz)
Vin in 0 PULSE(-1 1 0 1u 1u 5m 10m)
R1 in n1 5.1k
R2 n1 n2 5.1k
C1 out n1 100n
C2 n2 0 10n
Eop out 0 n2 out 1Meg
.tran 0 20m 0 2u
.end
```

Listing 3: Underdamped netlist

<Insert LTSpice transient plot: underdamped>

Figure 4: LTSpice transient response, underdamped case.

Measurements

We captured the input and output waveforms on the oscilloscope for each case. Table ?? compares the theoretical parameters against the values read off the scope, and the scope captures follow.

Table 2: Theoretical vs. measured parameters.

Case	Quantity	Theoretical	Measured	% Error
Critically damped	f_0 (Hz)	994.7	< >	< >
	settling time (μ s)	≈ 800	< >	< >
Overdamped	f_0 (Hz)	981.0	< >	< >
	slow τ (μ s)	470	< >	< >
Underdamped	f_d (Hz)	936	< >	< >
	overshoot (%)	35.1	< >	< >
	t_p (μ s)	534	< >	< >

<Insert scope capture: critically damped, CH1 input / CH2 output>

Figure 5: Measured response, critically damped case.

<Insert scope capture: overdamped, CH1 input / CH2 output>

Figure 6: Measured response, overdamped case.

<Insert scope capture: underdamped, CH1 input / CH2 output>

Figure 7: Measured response, underdamped case.

Observations & Discussion

The three circuits behaved the way the theory predicts. The overdamped output climbed slowly to its final value with no overshoot, the critically damped output reached the final value quickly and just barely avoided overshoot, and the underdamped output overshoot and rang before settling. The simulation traces matched the hand-solved responses closely, since the ideal buffer in LTSpice has none of the limitations of the real op-amp.

The main differences between theory and measurement come from component tolerances and the real op-amp. The resistors and capacitors carry tolerances of several percent, and because Q and f_0 depend on ratios and products of these values, a small spread in the capacitors shifts the measured Q and the measured resonant frequency. The 741 also has a finite gain-bandwidth product and a limited slew rate, so on the fast edges of the underdamped case the buffer cannot follow the output

instantly, which slightly reduces the measured overshoot and smooths the corners compared with the ideal simulation. Breadboard parasitics and the loading of the scope probes add smaller shifts. < Add the specific measured percent differences once the bench data is filled in. >

If we did this again, the first change would be to measure each resistor and capacitor with the LCR meter before building and recompute α , ω_0 , and Q from the measured values, so the theoretical curve is based on the parts actually used rather than the nominal ones. Using 1% resistors and film capacitors would tighten the spread further. Replacing the 741 with a faster op-amp would remove most of the slew-rate rounding on the underdamped edges, and averaging the scope trace would clean up the noise on the overshoot and ringing measurements.